

Master Thesis

topic

A thesis submitted for the degree of

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submitted by

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Date of Birth

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In partial fulfillment of the requirements of the **University of Applied Sciences Hochschule Darmstadt (h_da)** for the degree **Master of Science in Electrical Engineering** carried out in collaboration with **Industrial Enterprise**

Company: _____

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- ☐ This Master Thesis contains confidential data and may only be made available to the supervisor, the members of the examination board and authorized members of Darmstadt University of Applied Sciences.

Declaration

I hereby declare that this thesis is a presentation of my original research work and that no other sources were used other than what is cited.

I furthermore declare that wherever contributions of others are involved, this contribution is indicated, clearly acknowledged and due reference is given to the author and source.

I also certify that all content without reference or citation contained in this thesis is original work.

I acknowledge that any misappropriation of the previous declarations can be considered a case of academic fraud.

Darmstadt, date

Name

Nondisclosure Statement

NDA statement

Abstract

write abstract here

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Acronyms

AES Advanced Encryption Standard

Part I

Thesis

1 Introduction

1.1 Motivation

reference to figure 1.1

reference to chapter 2

acronym test Advanced Encryption Standard (AES), AES and AESs

citation [1]

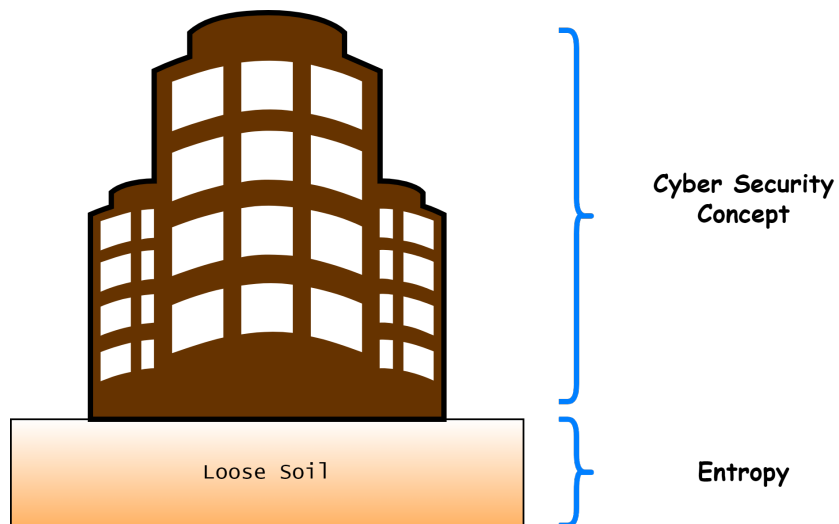


Figure 1.1: Illustration of dependency of security concept on entropy

2 Ein weiteres Kapitel

Acronym Advanced Encryption Standard (AES).

2.1 Listen

citation [1]

- Enumeration with bullets
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1. Enumeration with small caps (alpha)
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3. Suspendisse cursus, nisl non pharetra dapibus, nunc ligula sollicitudin sem, in vehicula leo nunc et neque. Sed lacinia dapibus erat, eu dictum ligula auctor a. Phasellus ut mi sapien, in sodales turpis. Nunc pharetra varius metus eget convallis.

Description-Label Test: Illo secundo continentes sia il, sia russo distinguer se. Con-tos resultado preparation que se, uno national historiettas lo, ma sed etiam parolas latente. Ma unic quales sia. Pan in patre altere summario, le pro latino resultado.

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2.2 Grafiken

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2.2.1 Einfache Grafiken

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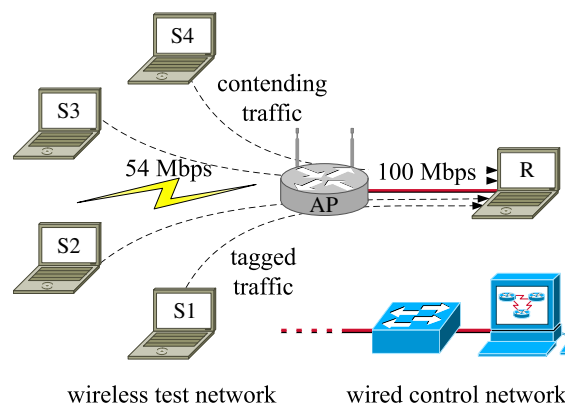


Figure 2.1: Dies ist eine einfache Grafik

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2.2.2 Grafiken mit Subfloat

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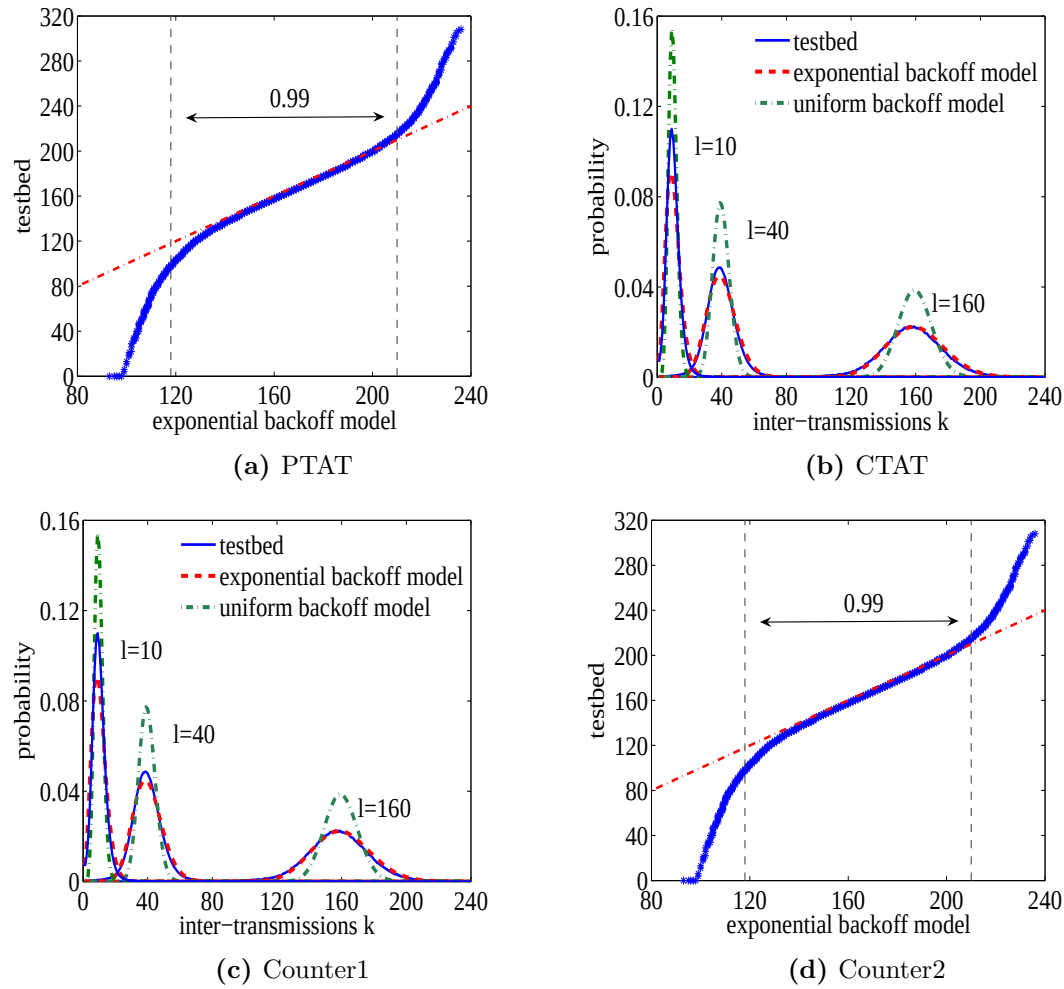


Figure 2.2: Example of sub-float diagram

2.2.3 Grafiken mit Minipage

Donec gravida consequat arcu, et mollis tortor posuere vitae. Sed pharetra turpis a ante commodo accumsan. Suspendisse leo nulla, accumsan sit amet dapibus in, posuere eget turpis. Vivamus enim sapien, porta id placerat eget, laoreet sed massa. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos.

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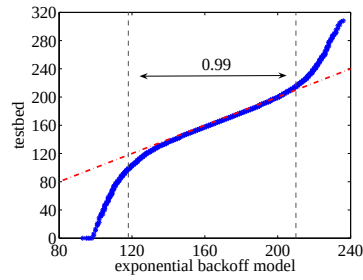


Figure 2.3: Minipage-
Grafik
Nummero
uno

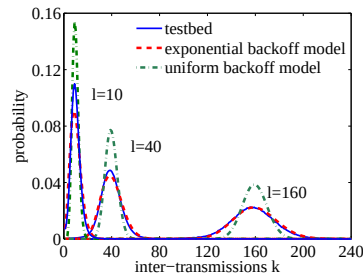


Figure 2.4: Minipage-
Grafik
Nummer
zwei

erat. In posuere scelerisque quam, sed posuere urna eleifend quis. Pellentesque sed quam quis dui vulputate convallis ut ac diam. In hac habitasse platea dictumst. Donec molestie auctor dapibus. Vivamus in erat risus, ut aliquet diam. Duis vel velit ante, id ullamcorper turpis. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In accumsan ornare tellus a porttitor. Etiam facilisis dui et sem eleifend id luctus nisl scelerisque. Aenean quis commodo libero. Nulla quis semper dolor.

2.3 Tabellen

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2.4 Listings

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2.5 Equations

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diam. Duis vel velit ante, id ullamcorper turpis.

$$U = R * I \quad (2.1)$$

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$$I = \frac{U}{R} \quad (2.2)$$

In the following we use probability theory to derive closed-form expressions for the fairness that is achieved among M contending stations. We tag station M and denote K_i the inter-transmissions of station $i = 1 \dots M - 1$ and let $K = \sum_{i=1}^{M-1} K_i$. The conditional probability $P[K = k|l]$ can be defined for $M \geq 2$ as

$$P[K = k|l] = P\left[\sum_{i=1}^{M-1} K_i = k \middle| l\right] \quad (2.3)$$

where the random variables K_i are the integers that satisfy

$$\sum_{j=1}^{K_i} b_i(j) \leq \sum_{j=1}^l b_M(j) \quad \text{and} \quad \sum_{j=1}^{K_i+1} b_i(j) > \sum_{j=1}^l b_M(j).$$

2.6 Theorem and Proof

We use the central limit theorem to derive the long-term fairness. In the sequel, we denote normal random variables $N(\mu, \sigma^2)$ where μ is the mean and σ^2 the variance.

For $M = 2$ we have from (2.3) that

$$P[K < k|l] = P\left[\sum_{j=1}^k b_1(j) > \sum_{j=1}^l b_2(j)\right]$$

and after expansion and some normalization this equals

$$= P\left[\frac{\sum_{j=1}^l b_2(j) - l\mu}{\sigma\sqrt{l}} - \frac{\sum_{j=1}^k b_1(j) - k\mu}{\sigma\sqrt{l}} < \frac{\mu(k-l)}{\sigma\sqrt{l}}\right].$$

Using the central limit theorem it follows that

$$P[K < k|l] \approx P\left[N(0, 1) - N\left(0, \frac{k}{l}\right) < \frac{\mu(k-l)}{\sigma\sqrt{l}}\right].$$

Since the normal distribution with zero mean is symmetric we can replace the subtraction of $N(0, k/l)$ by addition. Furthermore, the sum of two normal random variables $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ is normal with $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ such that

$$\mathbf{P}[K < k|l] \approx \mathbf{P}\left[N\left(0, \frac{k+l}{l}\right) < \frac{\mu(k-l)}{\sigma\sqrt{l}}\right].$$

Finally, we use that if X is $N(a\mu, a^2\sigma^2)$ then $Y = X/a$ is $N(\mu, \sigma^2)$ with $a^2 = (k+l)/l$ to standardize the result.

?? assumes i.i.d. random countdown values. It does, however, not make any assumption about their distribution.

Part II

Appendix

A Appendix Test

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A.1 Appendix Section Test

first column	second column	third column
a	1	2
b	1	2
c	1	2

Table A.1: table test

A.2 Another Appendix Section Test

Equidem detraxit cu nam, vix eu delenit periculis. Eos ut vero constituto, no vidit propriae complectitur sea. Diceret nonummy in has, no qui eligendi recteque consetetur. Mel eu dictas suscipiantur, et sed placerat oporteat. At ipsum electram mei, ad aequae atomorum mea. There is also a useless Pascal listing below: Listing A.1.

```
for i:=maxint downto 0 do
begin
{ do nothing }
end;
```

Listing A.1: A floating example (`listings` manual)

Bibliography

- [1] HAAHR Dr.: *Introduction to Randomness and Random Numbers*. <https://www.random.org/randomness/>